

FORMULA & CONCEPT SHEET

Boundary layer:-

→ Flow inside boundary layer is viscous 2D flow
 $v_x = f(x, y)$

→ $v_x = U_\infty$ (Inviscid flow outside boundary layer)

→ $v_x = 0.99 U_\infty$ (at $y = \delta$)

Approximation inside boundary layer:-

$$v_x \frac{\partial v_x}{\partial x} + v_y \frac{\partial v_x}{\partial y} = \nu \left(\frac{\partial^2 v_x}{\partial x^2} + \frac{\partial^2 v_x}{\partial y^2} \right)$$

R.C. $\left\{ \begin{array}{l} y=0 \Rightarrow v_x = v_y = 0 \\ y=\delta \Rightarrow v_x = U_\infty \\ x=0 \Rightarrow v_x = v \end{array} \right\}$ U_∞ & v might be different if plate is not flat. U_∞ may be a function of x .

There are 2 methods of finding boundary layer thickness:-

BLASIUS

MOMENTUM INTEGRAL EQ'n.

We define 2 thicknesses which may substitute boundary layer thickness in certain situations, where the desired quantity might be mass, momentum etc.

(I) Displacement Thickness (δ^*) = $\int_0^\delta (1 - \frac{u}{U_\infty}) dy$

If we want the same reduction in mass flow rate for an inviscid region then we can move the plate up by δ^* .

Means, above δ^* we have inviscid flow.

(II) Momentum Thickness (θ):
$$\int_0^{\delta} \frac{u}{U_{\infty}} \left(1 - \frac{u}{U_{\infty}}\right) dy$$

If we want the same reduction in momentum for an inviscid region then we can move the plate up by θ .

BLASIUS SOLUTION

If we notice, every where we are getting a term of the form $\left(\frac{v_x}{U_{\infty}}\right)$. Especially in δ^* & θ .

we assume
$$\frac{v_x}{U_{\infty}} = f(\eta) \quad ; \quad \eta = \frac{y}{\delta}$$

This is substituted in the governing eq'n we had got after applying boundary layer approximations.

We do this analysis near the boundary layer

$$v_x \propto U, \quad y \propto \delta, \quad \frac{\partial v_x}{\partial y} \approx 0$$

We arrive at an APPROXIMATE expression for δ

$$\delta = \sqrt{\frac{\nu x}{U}} \quad \Rightarrow \quad \eta = \frac{y}{\delta} = y \sqrt{\frac{U}{\nu x}}$$

We get

$$v_x = U \frac{df}{d\eta}, \quad v_y = \frac{1}{\sqrt{\frac{\nu U}{x}}} \left[\eta \frac{df}{d\eta} - f \right]$$

$$\frac{d\eta}{d\eta} = \frac{1}{2} \sqrt{\frac{x}{\nu}} \left[\frac{d^2 f}{d\eta^2} \right]$$

→ For numerical sol'n we apply some boundary conditions for η :

$$\begin{array}{l|l} y=0 \Rightarrow v_x = v_y = 0 \Rightarrow \eta = 0 & \eta = 0 \Rightarrow f = f' = 0 \\ v_x = U \Rightarrow \eta \rightarrow \infty & \eta = \infty \Rightarrow f' = 1 \end{array}$$

At $\eta = 5$ we get $\frac{df}{d\eta} = \frac{v_x}{U} = \underline{\underline{0.99155}}$ } Boundary layer